

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458305.67187 +/- 0.00086 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	1e-05 +/- 0.00047
Transit depth (Rp/Rs)^2	2.6e-09 +/- 4.9e-07 [%]
Orbital Inclination (inc)	85.69 +/- 1.77
Airmass coefficient 1 (a1)	26.02 +/- 21.78
Airmass coefficient 2 (a2)	0.854868 +/- 1.1e-05
Scatter in the residuals of the lightcurve fit is	59.81 %
Best Comparison Star	#2 - [501, 157]
Optimal Method	PSF photometry
Transit Duration (day)	0.0457 +/- 0.0044

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	nan $\pm$ nan vs 0.13031 (deviation nan σ)
Duration fit vs published	0.0457 d vs 0.0954 d (deviation 0.02 σ)
Mid-time O-C	0.0 min
Depth SNR	nan
Residual scatter	59.81 %

Light curve

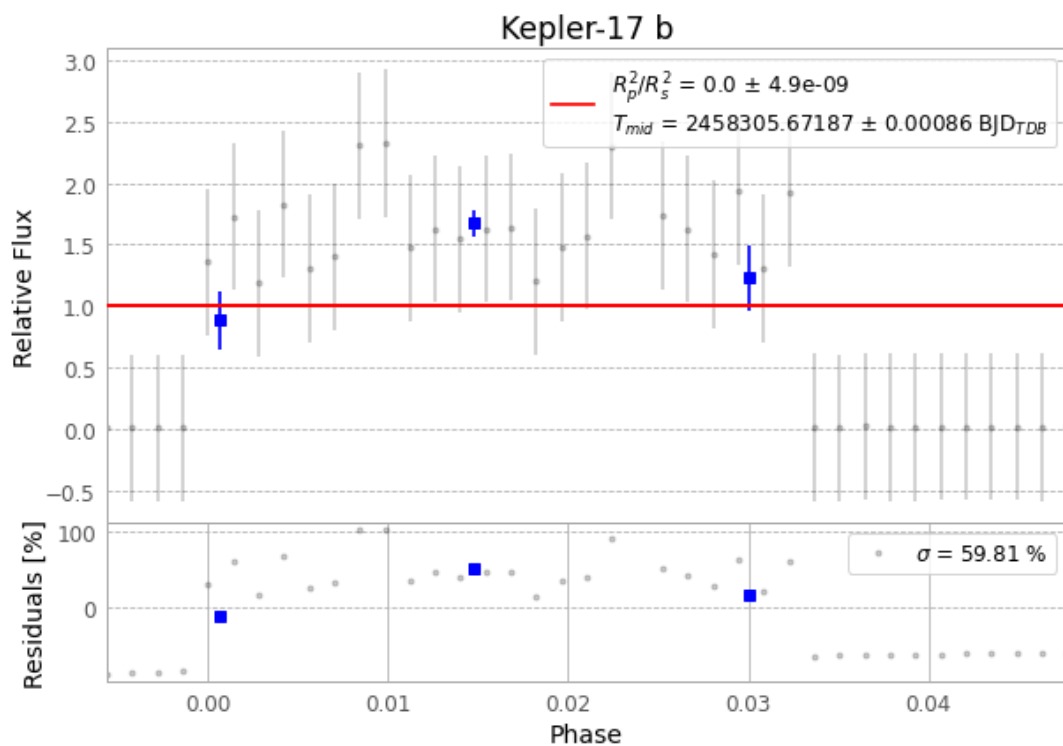


Figure 1 — FinalLightCurve_Kepler-17 b_2018-07-05.png (SHA-256 d0e73e9c07cfc932...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [375, 206], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a8d84a31190370b1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

39 FITS frames (26 MB), KEPLER-17180706035112.FITS → KEPLER-17180706054512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-KEPLER-17-180706.log (d9883438f0ca...), FinalLightCurve_Kepler-17 b_2018-07-05.png (d0e73e9c07cf...), FinalLightCurve_Kepler-17 b_2018-07-05.csv (32d1ec507a8c...)

Compute resources

Wall time 138.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-20 00:52Z.